APPLE IWORK & MAIL

CHEAT SHEETS

Pages, Numbers, Keynote & Mail — Shortcuts, Formulas and Automation

iCloud Collaboration · Versions · Templates · Styles · Page Layout · Table of Contents · Mail Merge · Track Changes · Numbers Tables · Header References · Pivot Tables · XLOOKUP · REGEX · Durations · Master Slides · Magic Move · Presenter Display · Smart Mailboxes · Mail Rules · AppleScript · Shortcuts · Writing Tools

28 REFERENCE SHEETS

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Introduction to iWork

APPLE'S FREE OFFICE SUITE

Pages (documents), **Numbers** (spreadsheets) and **Keynote** (presentations) come free on every Mac, iPad and iPhone, and run in a browser at **iCloud.com** on any computer. Files sync through **iCloud Drive**. They open and export Word, Excel and PowerPoint files. **Apple Creator Studio** (announced January 2026) is a subscription that adds premium templates, content and some intelligent features on top of the free apps.

NOTES

- App Store updates keep the apps current; check App Store › Updates
- iCloud.com versions need only an Apple Account, so PC users can collaborate
- Open Office files directly; export via File › Export To
- Each app has a **template chooser** on launch (turn it off in Settings)
- Creator Studio features are marked in the template chooser and menus

OPEN THINGS FAST

```
⌘N              # new document (template chooser)
⌘O              # open
⌘S  ⇧⌘S         # save / duplicate (save as)
⌘P              # print / PDF
File › Export To › Word / Excel / PowerPoint / PDF / EPUB
⇧⌘/             # Help menu search (find any command)
⌘,              # Settings
⌘W  ⌘Q          # close / quit

open -a Numbers budget.xlsx        # Terminal: open an Excel file in Numbers
```

CODE EXPLAINED

- → iWork autosaves; **Save** names a file the first time, after that it saves versions
- → Duplicate (`⇧⌘S`) replaces Save As; hold `⌥` in the File menu to see Save As
- → Shortcuts on this set are Mac shortcuts; iPad keyboard shortcuts mostly match
- → The Help menu search box finds and highlights any menu command

OTHER SUITES

- ⇄ **Microsoft 365** — Word / Excel / PowerPoint, desktop + web, subscription
- ⇄ **Google Workspace** — Docs / Sheets / Slides in the browser, free account

PLATFORM DIFFERENCES

- ▣ **Mac** — full feature set, AppleScript, Shortcuts
- ▣ **iPad / iPhone** — nearly full editing, Apple Pencil, touch layouts
- ▣ **iCloud (web)** — any browser; fewer features, good for collaboration

FILE FORMATS

FORMAT	NOTES
`.pages / .numbers / .key`	Native packages
`.docx / .xlsx / .pptx`	Open and export
`PDF`	Export, keeps layout
`EPUB`	Pages export (e-book)
`CSV / TSV`	Numbers import / export
`Movie / GIF`	Keynote export

WHEN IWORK FITS

GOOD FIT	LESS SO
`Design-heavy docs and flyers`	Huge data models
`Beautiful presentations`	Macro-driven workflows
`Small, readable spreadsheets`	Power Query / pivots at scale
`Apple-device teams`	Heavy Office round-tripping
`Free, no subscription`	Corporate admin controls

⚠ COMMON MISTAKES

- Sending `.pages` files to Windows users; export to Word or PDF
- Expecting Excel macros or VBA to survive import (they're dropped)
- Editing an exported copy and the original separately, so versions drift

✓ TIPS & BEST PRACTICES

- Use the Help menu search (`⇧⌘/`) whenever you can't find a command
- Keep work in iCloud Drive for versions and collaboration
- Turn off the template chooser if you always start blank

QUICK SUMMARY

- `Pages · Numbers · Keynote` — free on Apple devices
- `iCloud.com` — web versions
- `File › Export To` — Office / PDF
- `⇧⌘/` — Help menu search

Interface & Shortcuts That Work Everywhere

ONE LAYOUT ACROSS THREE APPS

All three apps share a layout: a **toolbar** across the top and a **sidebar** on the right with **Format** (style the selection), **Document** (page setup, Pages / Numbers) and **Animate** (Keynote). Shared shortcuts include **Copy Style / Paste Style** (copy formatting), **Paste and Match Style** (paste as plain text) and the standard Mac text shortcuts.

NOTES

- Customize the toolbar: right-click it › Customize Toolbar
- Format sidebar tabs change with the selection: Text, Arrange, Style, Table, Cell…
- `⌘T` Show Fonts and `⇧⌘C` Show Colors open the system panels
- Add your own shortcut for any menu item: System Settings › Keyboard › Keyboard Shortcuts › App Shortcuts
- View › Show Rulers (`⌘R`) for tabs and indents in Pages and Keynote

SHARED SHORTCUTS

```
⌥⌘C  ⌥⌘V       # copy style / paste style
⌥⇧⌘V           # paste and match style (plain text)
⌘B ⌘I ⌘U       # bold, italic, underline
⌘+  ⌘–         # bigger / smaller font
⌘{  ⌘|  ⌘}     # align left / center / right
^⌘+  ^⌘–       # superscript / subscript
⌘T  ⇧⌘C        # Fonts / Colors panels
⌘F  ⌘G  ⌘E     # find / find next / use selection for find
⌘K             # add link
⇧⌘K            # add comment
⌘R             # show / hide rulers
⇧⌘/            # Help menu search
```

CODE EXPLAINED

- → Copy Style copies character and paragraph formatting; it works across the three apps
- → Paste and Match Style takes on the formatting where you paste
- → `⌘E` loads the selected text into Find so `⌘G` jumps to the next match
- → Custom App Shortcuts need the exact menu item name, including "…"

OTHER SUITES

- ⇄ **Microsoft 365** — Format Painter, Paste Special `^⌘V` / `Ctrl+Alt+V`, ribbon
- ⇄ **Google Workspace** — paint format, `⌘⇧V` paste plain, menu search `⌥/`

PLATFORM DIFFERENCES

- ▣ **Mac** — every shortcut, custom App Shortcuts
- ▣ **iPad / iPhone** — hold `⌘` with a keyboard to see shortcuts; Format button (brush)
- ▣ **iCloud (web)** — Format sidebar and most shortcuts in the browser

SIDEBARS

SIDEBAR	HOLDS
Format	Text, shape, image, table, cell, chart styling
Document	Page size, margins, headers / footers (Pages)
Organize (Numbers)	Sort, filter, categories, pivot options
Animate (Keynote)	Transitions and builds
Arrange tab	Position, size, wrap, align

TEXT MOVEMENT

ACTION	KEYS
Word left / right	⌥← / ⌥→
Line start / end	⌘← / ⌘→
Doc start / end	⌘↑ / ⌘↓
Select by word	⌥⇧← / ⌥⇧→
Delete word	⌥⌫
Delete to line start	⌘⌫

⚠ COMMON MISTAKES

- Pasting web text with its fonts; use `⌥⇧⌘V`
- Restyling text by hand when Paste Style would copy it in one step
- Hunting menus for rare commands; type them into Help search

✓ TIPS & BEST PRACTICES

- Add toolbar buttons for commands you use daily
- Assign your own shortcut to any menu item you use a lot
- Learn the Format sidebar tabs; nearly every setting lives there

QUICK SUMMARY

- `⌥⌘C / ⌥⌘V` — copy / paste style
- `⌥⇧⌘V` — paste plain
- `Format sidebar` — styling
- `⇧⌘/` — find any command

Sheet 03

iCloud, Collaboration & Versions

iCloud Drive · Share › Collaborate · permissions & activity · Browse All Versions · password protection · export to Office / PDF

WORK TOGETHER AND GO BACK IN TIME

Save to **iCloud Drive** and you can **collaborate**: invite people by link, Messages or Mail, and edit at the same time from Mac, iPad, iPhone or iCloud.com. The **activity list** shows who changed what. **Revert To › Browse All Versions** brings back any earlier version, and a **password** protects sensitive files.

NOTES

- Share button (toolbar) › Collaborate › choose people and permission
- Permissions: Only invited people / Anyone with the link; Can make changes / View only
- Collaborators' cursors show in colour; the Collaboration button lists who's in
- File › Set Password to require a password to open
- Versions are kept automatically for files on iCloud Drive and the Mac

COLLABORATE AND RESTORE

1. Move the file to iCloud Drive (File › Move To)
2. Click **Share** › **Collaborate**; set Only invited people or Anyone with the link
3. Choose **Can make changes** or **View only**; send via Messages, Mail or copy link
4. Watch changes in the **activity list** (Collaborate button › Show All Activity)
5. To go back: File › **Revert To › Browse All Versions**
6. Pick a version and click **Restore**, or hold `⌥` and click Restore a Copy

WHERE TO FIND IT

TASK	WHERE
`Share / collaborate`	Share button
`Manage people`	Collaborate button › Manage
`Activity`	Collaborate button › Show All Activity
`Versions`	File › Revert To › Browse All Versions
`Password`	File › Set Password
`Export`	File › Export To

OTHER SUITES

⇄ **Microsoft 365** — OneDrive Share, AutoSave, Version History
⇄ **Google Workspace** — Share button, Version history (`⌘⌥⇧H`)

PLATFORM DIFFERENCES

- **Mac** — full sharing, Browse All Versions (Time Machine-style)
- **iPad / iPhone** — share and collaborate; versions on iCloud.com
- **iCloud (web)** — collaborate from any browser, Browse versions

PERMISSIONS

SETTING	MEANS
`Only invited people`	Must be on the list
`Anyone with the link`	Forwarding works too
`Can make changes`	Edit
`View only`	Read
`Allow others to invite`	Collaborators can add people

EXPORT OPTIONS

EXPORT	KEEPS
`Word / Excel / PowerPoint`	Most formatting
`PDF`	Exact layout, optional comments
`EPUB (Pages)`	Reflowable e-book
`CSV (Numbers)`	One table per file
`Movie / GIF (Keynote)`	Animations

⚠ COMMON MISTAKES

- Sharing "Anyone with the link" for private documents
- Exporting a copy to share instead of collaborating, then merging edits by hand
- Forgetting a password; Apple can't recover it

✓ TIPS & BEST PRACTICES

- Send iCloud links to PC users; they can edit in the browser
- Use Restore a Copy to compare an old version side by side
- Export PDFs for final copies so layout never shifts

QUICK SUMMARY

- `Share › Collaborate` — real-time editing
- `Activity list` — who changed what
- `Revert To › Browse All Versions` — restore
- `File › Set Password` — protect

Templates & Styles

SET UP ONCE, REUSE FOREVER

Every new file starts in the **template chooser**. Save your own with **File › Save as Template** and it appears under **My Templates**. Inside a file, **paragraph, character and list styles** (Pages, Keynote text) and **object styles** (shapes, tables, charts, text boxes) let you restyle many items at once. Creator Studio subscribers get extra premium templates.

NOTES

- Paragraph styles: Format sidebar › Text › style name; **Update** when you change one
- Style shortcuts: assign `F1` – `F8` from the paragraph style menu (Pages)
- Object styles: select an object › Format › Style › + to save its look
- Templates sync to other devices through iCloud
- Settings › General › For New Documents: choose a default template

AUTOMATE IT

- Save as Template + a default template = new files start correct
- Style shortcuts `F1` – `F8` for the styles you apply most

MAKE A TEMPLATE

1. Build the file: styles, colours, placeholders, logo
2. Turn text into placeholders: select text › Format › Advanced › Define as Placeholder Text
3. Turn images into media placeholders: Format › Image › Define as Media Placeholder
4. File › **Save as Template** › Add to Template Chooser
5. Name it; it shows under **My Templates**
6. Settings › General › For New Documents › Use template

WHERE TO FIND IT

TASK	WHERE
`Template chooser`	File › New (⌘N)
`Save template`	File › Save as Template
`Paragraph styles`	Format › Text › style menu
`Style shortcuts`	Style menu › Shortcuts (F1–F8)
`Object styles`	Format › Style › +
`Default template`	Settings › General

OTHER SUITES

⇄ **Microsoft 365** — `.dotx` / `.xltx` / `.potx`, Styles gallery, themes

⇄ **Google Workspace** — Template gallery, paragraph styles, `/copy` links

PLATFORM DIFFERENCES

- **Mac** — full template and style tools
- **iPad / iPhone** — use templates; create with Save as Template on iPad
- **iCloud (web)** — use templates; limited style editing

STYLE TYPES

TYPE	APPLIES TO
`Paragraph`	Whole paragraphs
`Character`	Selected words
`List`	Bullets / numbering
`Object`	Shapes, tables, charts, text boxes
`Cell (Numbers)`	Table cell styling via table styles

PLACEHOLDERS

KIND	BEHAVIOUR
`Placeholder text`	Replaced by typing
`Media placeholder`	Drop an image to replace
`Master placeholders (Keynote)`	Title, body, slide number
`Merge fields (Pages)`	Filled by mail merge
`Tags (Keynote)`	Placeholder identifiers for scripting

⚠ COMMON MISTAKES

- Changing one paragraph by hand and forgetting to click **Update**
- Copying last year's file instead of starting from a template
- Hard-coding colours instead of saving object styles

✓ TIPS & BEST PRACTICES

- Assign `F1` – `F8` to your most-used paragraph styles
- Save object styles for brand shapes and tables
- Keep templates in iCloud so every device has them

QUICK SUMMARY

- `Save as Template` — reuse a layout
- `Paragraph styles + Update` — change everywhere
- `F1-F8` — style shortcuts
- `Object styles` — reusable looks

Sheet 05

Word Processing vs Page Layout

document types & converting · sections · text boxes & linked text boxes · inline vs floating objects · text wrap · facing pages

TWO KINDS OF PAGES DOCUMENTS

Pages has two modes. **Word processing** documents flow text from page to page (reports, letters). **Page layout** documents are canvases where you place text boxes and images freely (flyers, newsletters). Switch with **File › Convert to Page Layout / Word Processing**. Objects can sit **inline** with text, **move with text**, or **stay on page**.

NOTES

- Document sidebar › Document › **Document Body** checkbox = word processing
- Linked text boxes (page layout): text flows from one box to the next
- Object placement: Format › Arrange › Stay on Page / Move with Text
- Text Wrap: Automatic, Around, Above and Below, Through, None
- Facing pages: Document › Document › Facing Pages for book layouts

LAYOUT SHORTCUTS

```
Insert › Page Break            # page break (word processing)
Insert › Section Break         # new section
Insert › Column Break          # next column
⇧⌘I            # show invisibles (¶, breaks)
⌘R             # show rulers
⌥⌘G  ⇧⌥⌘G      # group / ungroup objects
⇧⌘F  ⇧⌘B       # bring to front / send to back
⌥⇧⌘F ⌥⇧⌘B      # bring forward / send backward
⌘L   ⌥⌘L       # lock / unlock object
⌥ + drag       # duplicate object
```

CODE EXPLAINED

- → Show invisibles (`⇧⌘I`) reveals breaks and spaces so you can delete stray ones
- → Locked objects can't be moved by accident; unlock with `⌥⌘L`
- → Inline objects behave like big characters inside the text
- → Converting to Page Layout removes the main text body; content becomes text boxes

OTHER SUITES

- ⇄ **Microsoft 365** — Word for flowing docs; Publisher (retired) or PowerPoint for layouts
- ⇄ **Google Workspace** — Docs pageless / pages modes; no free page layout canvas

PLATFORM DIFFERENCES

- ▣ **Mac** — both modes, linked text boxes
- ▣ **iPad / iPhone** — both modes; touch arrange
- ▣ **iCloud (web)** — both modes, fewer layout controls

WHICH DOCUMENT TYPE?

USE	TYPE
Report, letter, essay, book	Word processing
Flyer, poster, menu	Page layout
Newsletter with columns	Either (columns or linked boxes)
Long manual with TOC	Word processing
Brochure / trifold	Page layout

OBJECT PLACEMENT

SETTING	BEHAVIOUR
Inline with text	Moves like a character
Move with text	Anchored to a paragraph
Stay on page	Fixed position
Wrap: Around	Text flows around
Wrap: None	Object sits over text

⚠ COMMON MISTAKES

- Building a flyer in word processing mode and fighting the text flow
- Spacing with blank lines and spaces instead of breaks and spacing
- Leaving images on Stay on Page in flowing documents, so they drift from their text

✓ TIPS & BEST PRACTICES

- Decide the type before you start; converting later rearranges content
- Turn on invisibles when layouts misbehave
- Lock background shapes so they don't move while editing

QUICK SUMMARY

- `Word processing` — text flows
- `Page layout` — free canvas
- `⇧⌘I` — show invisibles
- `Move with text / Stay on page` — object anchoring

Styles & Long Documents

REPORTS, BOOKS AND MANUALS IN PAGES

Long documents in Pages run on **paragraph styles**: apply Heading styles and the **table of contents** builds itself, both in the document and in the TOC sidebar for navigation. **Sections** let headers, footers and page numbering change, **footnotes** and **endnotes** number automatically, and **bookmarks** create cross-links inside the document.

NOTES

- TOC: Insert › Table of Contents › Document / Section; it updates automatically
- TOC sidebar: View › Table of Contents (click to jump)
- Headers / footers: Document sidebar › Section tab; Hide on first page
- Page numbers: click in header / footer › Insert › Page Number; Section tab › Start at
- Footnote: Insert › Footnote; switch to endnotes in Document › Footnotes

AUTOMATE IT

- Paragraph styles drive the TOC, so no manual page lists
- Bookmarks + links instead of typed "see page 12"

LONG-DOC SHORTCUTS

```
F1 … F8          # your paragraph style shortcuts
Insert › Page Break  # page break
⇧⌘I              # show invisibles
⌘K               # add link (to a bookmark, URL, email)
⌘F               # find; options for match case / whole words
⇧⌘W              # show word count

Insert › Table of Contents › Document
Insert › Footnote
Format › Bookmark (Bookmarks list in Document sidebar)
```

CODE EXPLAINED

- → Choose which styles appear in the TOC: select the TOC › Format sidebar › tick styles
- → The TOC updates as headings change; no refresh step
- → Sections start at section breaks; each has its own header / footer settings
- → Links to bookmarks keep working when text moves

OTHER SUITES

- ⇄ **Microsoft 365** — Word TOC + `F9`, captions & cross-references, section breaks
- ⇄ **Google Workspace** — Docs TOC (refresh manually), document tabs, footnotes

PLATFORM DIFFERENCES

- ▣ **Mac** — every option
- ▣ **iPad / iPhone** — TOC, footnotes, sections
- ▣ **iCloud (web)** — TOC and footnotes; fewer section options

SECTION OPTIONS

OPTION	EFFECT
`Hide on first page`	Clean cover page
`Left and right pages are different`	Book layout
`Match previous section`	Same header (off = new)
`Page numbering › Start at`	Restart numbering
`Section starts on right`	Chapters on odd pages

NOTES & REFERENCES

FEATURE	WHERE
`Footnote`	Insert › Footnote
`Endnotes / section endnotes`	Document › Footnotes type
`Bookmark`	Format › Bookmark
`Link to bookmark`	⌘K › Bookmark
`Citations (EndNote)`	Insert › EndNote Citation (with EndNote)

⚠ COMMON MISTAKES

- Headings made with bold Normal text; they never reach the TOC
- Changing a header and having every section change; turn off Match previous section
- Typing page numbers into the footer instead of inserting a page number field

✓ TIPS & BEST PRACTICES

- Set style shortcuts `F1` – `F8` for headings while writing
- Use the TOC sidebar as your navigation pane
- Export to PDF with "Include table of contents" for bookmarks in the PDF

QUICK SUMMARY

- `Paragraph styles` — power the TOC
- `Insert › Table of Contents` — auto-updating
- `Section tab` — headers / numbering per section
- `Format › Bookmark` — internal links

Sheet 07

Tables, Charts & Mail Merge

tables in Pages · charts · media placeholders · mail merge from Numbers / Contacts · merge fields · ePub & PDF export

DATA IN DOCUMENTS

Pages tables work like small Numbers tables, formulas included. Charts update when you edit their data. **Mail merge** fills **merge fields** from a **Numbers spreadsheet** or a **Contacts group** to make one personalised document per row, ready to print, PDF or share.

NOTES

- Table formulas: click a cell, type `=` (same functions as Numbers)
- Merge fields: Document sidebar › Document tab › **Mail Merge › Add Merge Field**
- Run the merge: Document button › **Mail Merge**, pick the Numbers file or Contacts group
- Preview with real data before merging
- Charts: Insert › Chart; Edit Chart Data opens the data editor

AUTOMATE IT

- Mail merge from a Numbers list instead of copying names into letters
- Drag a Contact card onto merge fields for a single quick letter

WHERE TO FIND IT

TASK	WHERE
`Add merge field`	Document › Mail Merge › Add Merge Field
`Mail merge`	Document button › Mail Merge
`Insert table`	Table button in toolbar
`Table formulas`	Type = in a cell
`Edit chart data`	Select chart › Edit Chart Data
`Export EPUB`	File › Export To › EPUB

RUN A MAIL MERGE

1. Keep recipients in Numbers: one header row, one row per person
2. In Pages, place the cursor; Document sidebar › Document › **Mail Merge › Add Merge Field**
3. Pick the field (e.g. First Name); repeat for each field
4. Click **Document** in the toolbar › **Mail Merge**
5. Choose the Numbers file (or Contacts group) under **Populate With**
6. Click **Preview**, then **Merge** to create the documents

OTHER SUITES

⇄ **Microsoft 365** — Word Mailings tab with an Excel list

⇄ **Google Workspace** — Docs mail merge (Workspace) or Apps Script

PLATFORM DIFFERENCES

- **Mac** — full mail merge (Numbers or Contacts)
- **iPad / iPhone** — mail merge from Numbers or a Contacts group
- **iCloud (web)** — merge fields limited

MERGE DATA SOURCES

SOURCE	BEST FOR
`Numbers spreadsheet`	Many recipients, custom fields
`Contacts group`	Address lists you already keep
`Single Contact card`	One quick letter (drag it in)
`Excel file`	Open in Numbers, save, then merge

TABLE BASICS IN PAGES

ACTION	KEYS / WHERE
`Next / previous cell`	Tab / ⇧Tab
`Add row below`	⌥↓
`Add column right`	⌥→
`Header rows`	Format › Table › Headers
`Repeat header on pages`	Format › Table › Repeat header rows
`Formula`	= in a cell

⚠ COMMON MISTAKES

- Blank rows or merged cells in the Numbers list
- Header names in Numbers that don't match the merge fields you picked
- Merging without previewing a few records

✓ TIPS & BEST PRACTICES

- Keep one master contacts spreadsheet and reuse it for every merge
- Use a Contacts group for holiday cards and labels
- Export the merged result to PDF for printing or emailing

QUICK SUMMARY

- `Add Merge Field` — placeholders
- `Document › Mail Merge` — run it
- `Numbers / Contacts` — data sources
- `Preview` — check before merging

Review & Editing

EDIT, REVIEW, PROOFREAD

Track Changes marks every edit with the author's colour so changes can be accepted or rejected. **Comments** and **highlights** hold notes without changing text. **Find & Replace** has options for case and whole words. macOS **Text Replacements**, **Dictation** and the Apple Intelligence **Writing Tools** speed up writing and proofreading.

NOTES

- Track Changes: Edit › Track Changes; toolbar for Accept / Reject
- Comment: `⇧⌘K`; highlight: `⇧⌘H`
- Find & Replace: `⌘F`, then the gear for Find & Replace and options
- Text Replacements: System Settings › Keyboard › Text Replacements (all apps)
- Writing Tools: select text, right-click › Writing Tools (Apple Intelligence Macs)

REVIEW SHORTCUTS

```
⇧⌘E            # Track Changes on / off
⇧⌘K            # add comment
⇧⌘H            # highlight
⌘F             # find (gear › Find & Replace)
⌘G  ⇧⌘G        # next / previous match
⌘E             # use selection for find
⇧⌘W            # show word count
⌘;             # next misspelled word
⇧⌘;            # spelling and grammar panel
fn fn / Globe twice  # start dictation
```

CODE EXPLAINED

→ Turn tracking on before editing; earlier edits aren't tracked

→ Comments and highlights can be replied to and resolved in the Comments pane

→ Text Replacements sync via iCloud to iPhone and iPad

→ Writing Tools can proofread, rewrite in a tone, or summarise selected text

OTHER SUITES

⇄ **Microsoft 365** — Word Track Changes `⇧⌘E` / `Ctrl+Shift+E`, Compare, Editor

⇄ **Google Workspace** — Docs Suggesting mode, Compare documents, regex replace

PLATFORM DIFFERENCES

- **Mac** — tracking, comments, Writing Tools (Apple Intelligence Macs)
- **iPad / iPhone** — tracking, comments, dictation, Writing Tools on supported devices
- **iCloud (web)** — comments; Track Changes viewing

TRACKING CONTROLS

CONTROL	DOES
`Accept / Reject`	Current change
`Accept All / Reject All`	Whole document
`Markup view`	Show changes inline
`Final view`	Hide markup
`Pause tracking`	Edit without tracking

WRITING TOOLS

TOOL	DOES
`Proofread`	Grammar and spelling fixes
`Rewrite / Friendly / Professional / Concise`	Tone changes
`Summary / Key Points`	Condense selection
`Table / List`	Restructure text
`Compose (ChatGPT ext.)`	Generate text, optional

⚠ COMMON MISTAKES

- Sending a file with tracked changes still visible; Accept All or export final
- Relying only on spell check; Proofread or read aloud catches more
- Creating Text Replacements that are real words

✓ TIPS & BEST PRACTICES

- Use highlights for "check this" notes that aren't full comments
- Read aloud: Edit › Speech › Start Speaking for a final proof
- Review Writing Tools changes before accepting them

QUICK SUMMARY

- `⇧⌘E` — Track Changes
- `⇧⌘K` / `⇧⌘H` — comment / highlight
- `Text Replacements` — system-wide expansion
- `Writing Tools` — proofread, rewrite

The Numbers Canvas Model

sheets vs tables · many tables per sheet · header rows / columns · footer rows · table names · vs an Excel grid

TABLES ON A BLANK PAGE

A Numbers **sheet** is a blank canvas, not an endless grid. On it you place **tables**, **charts**, text and images, each a separate object you can move and resize. Every table has its own **header rows**, **header columns** and **footer rows**, its own name, and only as many rows and columns as you give it. Formulas refer to tables by name.

NOTES

- Add a table: Table button in the toolbar; pick a style
- Rename: click the table name (turn it on in Format › Table › Table Name)
- Header rows stay visible when scrolling and can repeat on printed pages
- Footer rows are good for totals; they stay at the bottom as rows are added
- Resize a table by dragging the round handle at the bottom right

TABLE NAVIGATION & STRUCTURE

```
Tab / ⇧Tab        # next / previous cell (Tab at end adds a column)
↩ / ⇧↩            # next / previous row (↩ at end of last row adds a row)
⌥↑  ⌥↓            # add row above / below
⌥←  ⌥→            # add column before / after
⌥⌘⌫               # delete selected rows or columns
⌘↑ ⌘↓ ⌘← ⌘→       # jump to table edge
⌥↩                # line break inside a cell
Esc               # stop editing, keep cell selected
```

CODE EXPLAINED

- → A table only has the rows you add, so size it to the data
- → Header rows and columns are included in formulas by name, not by number
- → Click the table once to select it as an object; again to select a cell
- → Use multiple small tables (Inputs, Data, Summary) instead of one huge grid

OTHER SUITES

- ⇄ **Microsoft 365** — Excel: one huge grid per sheet; tables (`Ctrl+T`) are ranges inside it
- ⇄ **Google Workspace** — Sheets: grid per tab like Excel; tables via Convert to table

PLATFORM DIFFERENCES

- ▣ **Mac** — full table tools
- ▣ **iPad / iPhone** — tap and drag handles; Form view for data entry (iPad / iPhone)
- ▣ **iCloud (web)** — tables and formulas; fewer table options

TABLE PARTS

PART	USE
`Header rows (up to 5)`	Column titles, frozen
`Header columns (up to 5)`	Row labels, frozen
`Footer rows (up to 5)`	Totals at the bottom
`Table name`	Used in formulas across tables
`Body rows`	Data

NUMBERS VS EXCEL

	NUMBERS	EXCEL
`Layout`	Free canvas	Fixed grid
`Tables per sheet`	Many	One grid (+ tables)
`Reference style`	Header names, `Table::Cell`	A1 addresses
`Big data`	Small to medium	Up to 1M rows

⚠ COMMON MISTAKES

- Building one giant table like an Excel sheet; split into named tables
- Leaving default names like "Table 1" before writing cross-table formulas
- Typing totals into a body row instead of a footer row, so sorting moves them

✓ TIPS & BEST PRACTICES

- Name every table before writing formulas
- Put inputs, data and results on separate tables on one sheet
- Use footer rows for totals so new data rows go above them

QUICK SUMMARY

- `Sheet` — canvas
- `Table` — object with its own rows
- `Header / footer rows` — titles and totals
- `⌥↓ / ⌥→` — add row / column

Sheet 10

Navigation, Entry & Cell Formats

autofill · data formats (number, currency, duration, date) · checkboxes · star ratings · sliders & steppers · pop-up menus

INTERACTIVE CELLS

Numbers cells can be more than text: **checkboxes**, **star ratings**, **sliders**, **steppers** and **pop-up menus** are all cell formats (Format › Cell › Data Format). **Autofill** extends patterns by dragging the yellow handle. **Duration** is a real format, so time tracking adds up correctly.

NOTES

- Autofill: select cells, drag the yellow dot on the edge
- Data Format pop-up: Automatic, Number, Currency, Percentage, Date & Time, Duration…
- Pop-up Menu format: list choices once, then pick in each cell
- Checkbox values are TRUE / FALSE in formulas
- Form (iPad / iPhone): enter one row at a time like a form

AUTOMATE IT

- Pop-up menus instead of typed status values, so sorting and counting stay clean
- Checkbox + COUNTIF(…, TRUE) for live progress counts

ENTRY SHORTCUTS

```
=                  # start a formula in the cell
⌥↩                 # line break in a cell
⌥⌘C   ⌥⌘V          # copy / paste style only
⇧⌘V                # paste formula results (values)
⌥⇧⌘V               # paste and match style
Table › Autofill Cells › Down    # fill down
Table › Autofill Cells › Right
Format › Cell › Data Format › Pop-Up Menu
⌥ + drag cell      # copy a cell
```

CODE EXPLAINED

→ Numbers has no "insert today" shortcut; use `=TODAY()` and paste results to freeze it

→ Paste Formula Results turns formulas into fixed values

→ Autofill continues series like Monday → Tuesday, Q1 → Q2, 1, 2 → 3

→ Pop-up menu options can include a blank first item for "not set"

OTHER SUITES

⇄ **Microsoft 365** — Excel Data Validation lists, Insert › Checkbox, Flash Fill

⇄ **Google Workspace** — dropdown chips, Insert › Checkbox, Smart Fill

PLATFORM DIFFERENCES

- **Mac** — every format and autofill
- **iPad / iPhone** — Form view, tap formats
- **iCloud (web)** — most formats; sliders / steppers limited

DATA FORMATS

FORMAT	USE
`Number / Currency / %`	Values
`Date & Time`	Dates, custom display
`Duration`	2h 30m that adds up
`Checkbox`	TRUE / FALSE
`Star Rating`	0–5 stars
`Slider / Stepper`	Bounded numeric input
`Pop-Up Menu`	Pick from a list

AUTOFILL SERIES

START	DRAG GIVES
`Monday`	Tuesday, Wednesday…
`Jan`	Feb, Mar…
`1, 2 (both selected)`	3, 4, 5…
`Q1`	Q2, Q3, Q4…
`Item 1`	Item 2, Item 3…

⚠ COMMON MISTAKES

- Typing "Done" / "done" / "DONE"; use a pop-up menu
- Storing times as text ("2:30") instead of a Duration format
- Expecting Excel's `Ctrl+;` today shortcut

✓ TIPS & BEST PRACTICES

- Use sliders or steppers for model inputs people will tweak
- Star ratings make quick scoring sheets
- Paste Formula Results to freeze calculated values

QUICK SUMMARY

- `Data Format` — checkboxes, menus, durations
- `Autofill handle` — extend series
- `⇧⌘V` — paste results
- `Pop-Up Menu` — consistent entries

Formula Basics

= formula editor · header-name references · Table 1::B2 · Sheet 1::Table 1::B2 · preserve row / column ($) · Functions Browser

FORMULAS THAT READ LIKE SENTENCES

Type = in a cell to open the **formula editor** and the **Functions Browser**. Numbers can refer to cells by **header names** (`Price Widget` = Price column, Widget row) as well as by address. Other tables use `Table::Cell`, other sheets `Sheet::Table::Cell`. Click a reference's arrow to **preserve** row or column, the equivalent of $.

NOTES

- Header-name references need header rows / columns turned on
- Click cells or drag ranges while editing a formula; Numbers writes the reference
- Preserve: click the reference token's triangle › Preserve Row / Preserve Column
- Quick calculations: select cells and read Sum, Average, Min, Max at the bottom
- Errors show a red triangle; click it for the message

REFERENCES & OPERATORS

```
=B2×C2                          # Numbers shows × for *
=Price × Quantity               # header names (same row)
=SUM(Amount)                    # whole Amount column
=SUM(Sales::Amount)             # column in another table
=Inputs::B2                     # cell in table "Inputs"
=Summary::Totals::B2            # sheet::table::cell
=B2×$F$1                        # preserved row and column
="Total: "&D2                   # join text with &
=A2<>""                         # not empty
=(B2-C2)÷C2                     # growth
```

CODE EXPLAINED

- → Numbers displays `*` as `×` and `/` as `÷`; type either
- → A whole-column reference like `SUM(Amount)` excludes header and footer rows
- → Cross-table references update if you rename the table
- → The Functions Browser (right side) shows syntax and examples for every function

OTHER SUITES

- ⇄ **Microsoft 365** — A1 refs with `$`, `Sheet!A1`, Name Manager for names
- ⇄ **Google Workspace** — A1 refs with `$`, `'Tab'!A1`, named ranges, open ranges `A2:A`

PLATFORM DIFFERENCES

- ▣ **Mac** — formula editor, Functions Browser
- ▣ **iPad / iPhone** — formula keyboard with function list
- ▣ **iCloud (web)** — formulas and references; fewer browser examples

REFERENCE STYLES

STYLE	EXAMPLE
Address	B2, B2:B20
Whole column / row	B, 2:2
Header name	Amount, Price Widget
Other table	Sales::Amount
Other sheet	Q1::Sales::B2
Preserved	B2

COMMON ERRORS

MESSAGE	MEANS
Invalid reference	Deleted / renamed target
Can't divide by zero	Zero or blank divisor
Type mismatch	Text where a number was expected
Circular reference	Formula refers to itself
Syntax error	Missing parenthesis / argument

⚠ COMMON MISTAKES

- Copying formulas without preserving the input cell, so references drift
- Header names with spaces or duplicates that make references ambiguous
- Forgetting that renaming a table changes every formula's text (correctly, but surprisingly)

✓ TIPS & BEST PRACTICES

- Turn on header rows and use names; formulas explain themselves
- Keep inputs in their own table and refer to them by `Inputs::`
- Use quick calculations at the bottom instead of writing throwaway formulas

QUICK SUMMARY

- `=` — formula editor
- `Amount` — header-name reference
- `Table::Cell` — cross-table
- `Preserve row / column` — like `$`

Sort, Filter, Categories & Pivot Tables

ORGANIZE DATA WITHOUT FORMULAS

The **Organize** sidebar sorts and filters a table. **Categories** group rows by a column value with collapsible **summary rows** (sum, count, average), like subtotals that update as you edit. **Pivot tables** (Organize › Create Pivot Table) summarise a source table into rows, columns and values, and refresh when the data changes.

NOTES

- Organize button (toolbar) › Sort / Filter / Categories
- Categories by date can group by year, quarter, month or week
- Summary row: click the category header cell › pick Sum, Count, Average…
- Pivot: select a table › Organize › Create Pivot Table › On New Sheet / Current Sheet
- Pivot options: Rows, Columns, Values in the Organize sidebar; Refresh after source edits

AUTOMATE IT

- Categories with summary rows instead of hand-written subtotals
- A pivot table that refreshes in one click instead of a rebuilt summary

GROUP AND SUMMARISE

1. Select the table; click **Organize** in the toolbar
2. **Categories** › Add a Category › pick the column (e.g. Region)
3. Click a category row's cell under Amount › choose **Sum**
4. For a cross-tab: Organize › **Create Pivot Table**
5. Drag fields to **Rows**, **Columns** and **Values**
6. After editing the source table, click **Refresh** on the pivot

WHERE TO FIND IT

TASK	WHERE
Sort	Organize › Sort
Filter	Organize › Filter
Categories	Organize › Categories
Pivot table	Organize › Create Pivot Table
Refresh pivot	Pivot › Organize › Refresh
Quick filter	Click column header › Quick Filter

OTHER SUITES

⇄ **Microsoft 365** — Excel Sort & Filter, Subtotal, PivotTables (refresh needed)

⇄ **Google Workspace** — Sheets filters / filter views, pivot tables (live)

PLATFORM DIFFERENCES

- **Mac** — categories and pivot tables
- **iPad / iPhone** — sort, filter, categories, pivots
- **iCloud (web)** — sort and filter; pivot support limited

SUMMARY OPTIONS

FUNCTION	SHOWS
Sum	Total per group
Count	Rows per group
Average	Mean
Minimum / Maximum	Extremes
Count Unique	Distinct values (pivots)

FILTER RULES

RULE	EXAMPLE
Text is / contains	Region is West
Number greater than	Amount > 1,000
Date in the last	30 days
Top / bottom	Top 10
Match all / any	AND / OR rules

⚠ COMMON MISTAKES

- Forgetting to refresh pivot tables after editing the source
- Filtering a shared table and confusing collaborators (filters apply to everyone)
- Typing subtotals by hand that go stale

✓ TIPS & BEST PRACTICES

- Use categories for quick grouping; pivots for cross-tabs
- Turn off filters before sending a file
- GETPIVOTDATA pulls a pivot value into a formula

QUICK SUMMARY

- `Organize` — sort, filter, categories
- `Summary rows` — live subtotals
- `Create Pivot Table` — cross-tab
- `Refresh` — after source edits

Logic, Math & Lookup Functions

DECIDE, SUMMARISE, LOOK UP

Numbers supports the same core functions as Excel and Sheets: `IF`, `IFS`, `SWITCH`, the conditional `…IFS` family, and modern lookups like `XLOOKUP` and `XMATCH`. Header names make them readable: `=SUMIFS(Amount, Region, "West")`. Recent versions add array functions such as `FILTER`, `UNIQUE` and `SORT`.

NOTES

- Criteria: `">100"`, `"<>Closed"`, `">="&F1`
- Wildcards in criteria: `"*north*"`; Numbers also accepts `REGEX("…")` as a condition
- SUMIFS: sum values first, then test values / condition pairs
- `XLOOKUP(search-for, search-where, result-values, if-not-found, match-type, search-type)`
- Check the Functions Browser for argument names in your version

AUTOMATE IT

- One SUMIFS table driven by input cells instead of filtered copies
- XLOOKUP against a Prices table instead of retyping prices

LOGIC & LOOKUP FORMULAS

```
=IF(Score≥70, "Pass", "Fail")
=IFS(Score≥90,"A", Score≥80,"B", Score≥70,"C", TRUE,"F")
=SWITCH(Code, "N","North", "S","South", "Other")
=SUMIFS(Sales::Amount, Sales::Region, "West")
=COUNTIFS(Status, "Open", Due, "<"&TODAY())
=AVERAGEIFS(Amount, Product, "Widget")
=SUMIFS(Amount, Name, REGEX("^A"))                    # names starting with A
=XLOOKUP(SKU, Products::SKU, Products::Price, "Not found")
=XLOOKUP(Income, Rates::Min, Rates::Rate, , -1)       # bracket
=INDEX(Products::Price, MATCH(SKU, Products::SKU, 0))
=IFERROR(B2÷C2, 0)
=FILTER(Sales::A2:D100, Sales::Region="West")         # recent versions
```

CODE EXPLAINED

- → `≥` and `>=` both work; Numbers displays the symbol
- → `REGEX("…")` as a criteria value lets SUMIFS / COUNTIFS match patterns
- → XLOOKUP match-type `-1` = exact or next smaller
- → Array functions spill results into the cells below in recent versions

OTHER SUITES

- ⇄ **Microsoft 365** — same IF / IFS / SUMIFS / XLOOKUP; FILTER, GROUPBY
- ⇄ **Google Workspace** — same functions; QUERY for SQL-style summaries

PLATFORM DIFFERENCES

- ▣ **Mac / iPad / iPhone** — same functions in current versions
- ▣ **iCloud (web)** — most functions; check the browser for newer ones

CRITERIA EXAMPLES

CRITERIA	MATCHES
`"West"`	Exactly West
`">100"`	Over 100
`"<>"`	Not blank
`"*north*"`	Contains north
`REGEX("^\d{5}$")`	Five digits
`">="&F1`	At least F1

LOOKUP CHOICES

NEED	USE
`One value, exact`	XLOOKUP
`Bracket / tier`	XLOOKUP(…, , -1)
`Position`	XMATCH
`Older files`	INDEX / MATCH, VLOOKUP
`Every match`	FILTER (recent)

⚠ COMMON MISTAKES

- Putting the sum values last in SUMIFS
- Writing `">F1"` instead of `">"&F1`
- VLOOKUP with approximate matching on unsorted data

✓ TIPS & BEST PRACTICES

- Use header names in formulas for readability
- Use REGEX conditions instead of helper columns
- Wrap lookups with an if-not-found value instead of IFERROR

QUICK SUMMARY

- `IF / IFS / SWITCH` — choose
- `SUMIFS` — conditional sum
- `REGEX("…")` — pattern criteria
- `XLOOKUP` — modern lookup

Text, Date & Regex Functions

TEXT CLEAN-UP AND DATE MATH

Numbers has modern text functions (TEXTBEFORE, TEXTAFTER, TEXTBETWEEN, TEXTSPLIT, TEXTJOIN) and regular expressions through REGEX (as a condition inside other functions), REGEX.EXTRACT and COUNTMATCHES. Dates and **durations** are first-class: DATEDIF, EOMONTH, NETWORKDAYS, plus DUR2HOURS and friends to turn durations into numbers.

NOTES

- `REGEX.EXTRACT(source, regex, occurrence, capture-group)`
- `REGEX("pattern")` can be used where a condition or search string is expected (SUBSTITUTE, COUNTIF, MATCH…)
- Add `(?i)` to a pattern for case-insensitive matching
- Durations: format a cell as Duration; `DUR2HOURS(B2)` gives a number
- Date & Time format can show date, time or both

TEXT, REGEX & DATE FORMULAS

```
=TEXTBEFORE(Email, "@")                    # jane
=TEXTAFTER(Email, "@")                     # acme.com
=TEXTBETWEEN(A2, "(", ")")                 # text inside parentheses
=TEXTJOIN(", ", TRUE, B2:B10)
=PROPER(TRIM(A2))
=REGEX.EXTRACT(A2, "\d{3}-\d{4}")          # 555-1234
=REGEX.EXTRACT(A2, "(\w+)@(\w+)", 1, 2)     # 2nd capture group
=COUNTMATCHES(A2, REGEX("\b\w+\b"))        # word count
=SUBSTITUTE(A2, REGEX("\s+"), " ")          # collapse spaces
=EOMONTH(TODAY(), 0)
=DATEDIF(Birth, TODAY(), "Y")
=NETWORKDAYS(Start, End)
=DUR2HOURS(Worked)×Rate                    # duration × hourly rate
```

CODE EXPLAINED

- → Numbers patterns use ICU regular expressions; `\d`, `\w`, `\s`, groups and `(?i)` all work
- → REGEX inside SUBSTITUTE replaces every match
- → COUNTMATCHES counts how many times a pattern appears
- → Duration × number needs `DUR2HOURS` (or similar) first

OTHER SUITES

- ⇄ **Microsoft 365** — TEXTBEFORE / TEXTAFTER / TEXTSPLIT, REGEXTEST / REGEXEXTRACT / REGEXREPLACE
- ⇄ **Google Workspace** — SPLIT, REGEXMATCH / REGEXEXTRACT / REGEXREPLACE (RE2)

PLATFORM DIFFERENCES

- ▣ **Mac / iPad / iPhone** — same functions in current versions
- ▣ **iCloud (web)** — most text and date functions

TEXT FUNCTIONS

FUNCTION	DOES
TEXTBEFORE / TEXTAFTER	Part of text
TEXTBETWEEN	Between two markers
TEXTSPLIT	Split into cells
TEXTJOIN	Join with delimiter
REGEX / REGEX.EXTRACT	Patterns
COUNTMATCHES	Count occurrences
PLAINTEXT	Strip formatting

DATE & DURATION

FUNCTION	RETURNS
TODAY() / NOW()	Date / date + time
DATEDIF(a, b, "M")	Whole months
EOMONTH / EDATE	Month math
NETWORKDAYS / WORKDAY	Business days
DAYNAME / MONTHNAME	Monday / March
DURATION(w,d,h,m)	Build a duration
DUR2HOURS / DUR2MINUTES	Duration → number

⚠ COMMON MISTAKES

- Multiplying a duration by a rate directly; convert with DUR2HOURS
- Dates typed as text in an unusual format; set Date & Time format first
- Forgetting regex is case-sensitive; add `(?i)`

✓ TIPS & BEST PRACTICES

- Use TEXTBETWEEN for codes in brackets or quotes
- COUNTMATCHES + REGEX makes a quick word or tag counter
- Track time as Duration so totals read "12h 30m"

QUICK SUMMARY

- `TEXTBEFORE / AFTER / BETWEEN` — slice text
- `REGEX.EXTRACT` — pull matches
- `REGEX("…")` — pattern inside other functions
- `DUR2HOURS` — duration to number

Charts & Conditional Highlighting

SHOW IT AND FLAG IT

Select data and click **Chart** to add a chart that stays linked to the table. **Interactive charts** add a slider to step through data sets. **Conditional highlighting** (Format › Cell › Conditional Highlighting) colours cells that meet rules: values, text, dates, durations, blanks, or matching another cell.

NOTES

- Edit Data References: select the chart › Edit Data References, then drag the coloured ranges
- Plot rows or columns as series: toggle in the chart's data editing bar
- Rules can compare to a cell (e.g. greater than Target::B2)
- Copy rules with Copy Style / Paste Style (`⌥⌘C` / `⌥⌘V`)
- Numbers highlighting has no custom formula rules; use a helper column for complex logic

AUTOMATE IT

- Highlight rules on a whole column apply to new rows automatically
- Interactive chart for monthly dashboards instead of 12 separate charts

ADD A CHART AND A RULE

1. Select the cells to chart (include header cells)
2. Click **Chart** in the toolbar; pick 2D, 3D or Interactive
3. Style with the Format sidebar (Chart, Axis, Series tabs)
4. Select a column of cells › Format › Cell › **Conditional Highlighting**
5. Add a Rule: e.g. Numbers › Greater than › 1000 › Red Fill
6. For complex logic, add a helper column with a formula and highlight on TRUE

WHERE TO FIND IT

TASK	WHERE
`Add chart`	Chart button
`Change data`	Edit Data References
`Chart style`	Format › Chart
`Highlight rule`	Format › Cell › Conditional Highlighting
`Copy rule`	⌥⌘C / ⌥⌘V
`Show values on chart`	Format › Series › Value Labels

OTHER SUITES

- ⇄ **Microsoft 365** — Excel charts, conditional formatting with formula rules
- ⇄ **Google Workspace** — Sheets chart editor, SPARKLINE, custom formula rules

PLATFORM DIFFERENCES

- **Mac** — every chart and rule type
- **iPad / iPhone** — charts and rules
- **iCloud (web)** — charts and basic rules

CHART PICKS

SHOW	CHART
`Trend`	Line
`Compare categories`	Column / bar
`Share of total`	Stacked bar / pie
`Relationship`	Scatter / bubble
`Many periods, one view`	Interactive
`Two measures`	Mixed / 2-axis

HIGHLIGHT RULE TYPES

RULE	EXAMPLE
`Numbers`	Greater than 1,000
`Text`	Contains "urgent"
`Dates`	In the last 7 days
`Durations`	More than 8h
`Blank`	Is blank
`With a cell`	Less than Target::B2

⚠ COMMON MISTAKES

- Charting totals rows along with data, doubling the bars
- Expecting formula-based highlight rules; use a helper column
- 3D charts that hide values; prefer 2D

✓ TIPS & BEST PRACTICES

- Include header cells so series get names
- Use value labels instead of gridlines for small charts
- Compare against a target cell so changing the target updates highlights

QUICK SUMMARY

- `Chart button` — linked chart
- `Edit Data References` — change source
- `Conditional Highlighting` — rule colours
- `Helper column` — complex rules

Numbers ↔ Excel

LIVING IN A MIXED OFFICE

Numbers opens `.xlsx` files and exports them, and recent versions share most of Excel's modern functions (XLOOKUP, FILTER, UNIQUE, LET, LAMBDA, TEXTSPLIT). The gaps are in features, not formulas: no VBA, no Power Query, no data model, and layout differences because Numbers puts tables on a canvas. Know what survives a round trip before you send a file.

NOTES

- Import: open the `.xlsx`; Numbers shows a warning list of what changed
- Export: File › Export To › Excel; choose one sheet per table or one per sheet
- Each Numbers table becomes its own Excel sheet unless you combine them
- Macros (VBA) are removed on import
- Check the Functions Browser before relying on a function a coworker uses in Excel

ROUND-TRIP CHECKLIST

```
# Before importing an Excel file
1. Save a copy of the original .xlsx
2. Open in Numbers → read the import warnings
3. Check: pivot tables, charts, conditional formats, data validation

# Before exporting to Excel
File › Export To › Excel › Advanced Options:
   ☑ Include a summary worksheet       ☐ one worksheet per table
4. Open the export in Excel (or Excel for the web) and test key formulas
5. Replace Numbers-only features (pop-up menus, sliders, durations)

# Functions to double-check in both directions
GROUPBY · PIVOTBY · REGEXTEST · REGEXREPLACE · COPILOT · PY
DUR2HOURS · REGEX · COUNTMATCHES · TEXTBETWEEN · STOCK
```

CODE EXPLAINED

- → Excel-only functions (e.g. GROUPBY, PIVOTBY) become values or errors in Numbers
- → Numbers-only functions (DUR2HOURS, REGEX conditions, COUNTMATCHES) turn into values in Excel
- → Pop-up menus become data validation lists; sliders become plain numbers
- → Always reopen exports in the other app before sending them out

OTHER SUITES

- ⇄ **Microsoft 365** — Excel is the usual destination; check in Excel for the web
- ⇄ **Google Workspace** — Sheets opens `.xlsx` too; QUERY / IMPORTRANGE have no Numbers equivalent

PLATFORM DIFFERENCES

- ▣ **Mac** — import / export options in full
- ▣ **iPad / iPhone** — export to Excel from the Share menu
- ▣ **iCloud (web)** — download as Excel

USUALLY LOST OR CHANGED

EXCEL FEATURE	IN NUMBERS
VBA macros	Removed
Power Query / data model	Values only
Slicers / timelines	Not supported
Sparklines	Not supported
Some chart types	Converted
Excel-only functions	Values or errors

WHEN TO HAND OFF

SITUATION	USE
Coworkers live in Excel	Excel for the web via a link
Shared live editing with PC users	iCloud.com or Google Sheets
Data > ~100k rows	Excel / Sheets
Automation needed on a server	Excel + Office Scripts / Sheets + Apps Script
Design-heavy report	Numbers (export PDF)

⚠ COMMON MISTAKES

- Sending an exported file without opening it in Excel first
- Assuming every Excel function exists in Numbers (or the reverse)
- Editing both the .numbers and .xlsx copies so they drift apart

✓ TIPS & BEST PRACTICES

- Keep one master format; export only for sharing
- Use PDF for read-only sharing; layout is guaranteed
- For mixed teams, iCloud.com links let PC users edit in the browser

QUICK SUMMARY

- `Import warnings` — read them
- `Export To › Excel` — per sheet / table
- `No VBA / Power Query` — features, not formulas, differ
- `Reopen exports` — test before sending

Themes & Master Slides

DESIGN THE DECK ONCE

Every Keynote presentation uses a **theme** made of **master slides** (Title, Title & Bullets, Photo…). Edit the masters and every slide based on them updates. **Save Theme** adds your design to the theme chooser so new decks start on-brand. Creator Studio subscribers see additional premium themes.

NOTES

- Edit masters: View › Edit Master Slides (or right-click a slide › Edit Master Slide)
- Placeholders: Format › Style › Define as Text / Media Placeholder
- Change a slide's master: Format sidebar › Slide Layout › master thumbnail
- File › Save Theme › Add to Theme Chooser
- Change theme on an existing deck: Document sidebar › Change Theme

AUTOMATE IT

- Custom theme with master slides instead of copying old decks
- Media placeholders so dropped photos crop and position themselves

BUILD A CUSTOM THEME

1. Start from a simple theme; open **View › Edit Master Slides**
2. Set fonts and colours on the main text placeholders
3. Add the logo to each master you'll use; delete masters you won't
4. Add media placeholders where photos go
5. Click **Done**, then File › **Save Theme** › Add to Theme Chooser
6. Apply to old decks via Document sidebar › Change Theme

WHERE TO FIND IT

TASK	WHERE
Edit masters	View › Edit Master Slides
Change slide layout	Format › Slide Layout
Reapply master	Format › Slide Layout › Update
Save theme	File › Save Theme
Change theme	Document › Change Theme
Slide size	Document › Slide Size

OTHER SUITES

⇄ **Microsoft 365** — PowerPoint Slide Master, layouts, `.potx` templates

⇄ **Google Workspace** — Slides › Edit theme (theme builder), import theme

PLATFORM DIFFERENCES

- **Mac** — full master editing
- **iPad / iPhone** — edit masters, save themes
- **iCloud (web)** — apply masters; limited master editing

MASTERS WORTH KEEPING

MASTER	USE
Title	Opening slide
Section	Divider
Title & Bullets	Most content
Photo / Photo – Horizontal	Image slides
Statement / Big Fact	One message
Blank	Custom layouts

WHERE TO EDIT

EDIT ON	CHANGES
Master slide	All slides using it
Single slide	That slide only
Theme	Every master
Object style	Shapes / text boxes using it

⚠ COMMON MISTAKES

- Restyling each slide instead of the master
- Pasting logos per slide; put them on masters
- Typing into free text boxes, which don't follow master changes

✓ TIPS & BEST PRACTICES

- Name custom masters clearly for others
- Keep a "style guide" slide in the theme
- Use Update in Slide Layout after pasting slides from another deck

QUICK SUMMARY

- `Edit Master Slides` — deck-wide design
- `Save Theme` — reuse it
- `Media placeholders` — drop-in photos
- `Change Theme` — restyle old decks

Sheet 18

Building Slides & Animations

Magic Move · build in / build out · Action builds · Animate sidebar & build order · alignment guides · Instant Alpha · object styles

MOTION WITH MEANING

Magic Move is Keynote's signature transition: duplicate a slide, move or resize objects on the copy, and Keynote animates the change. **Builds** bring objects in (build in), out (build out) or animate them in place (**Action**). The **Build Order** window sets sequence and timing. **Instant Alpha** removes image backgrounds.

NOTES

- Animate button (toolbar) opens the Animate sidebar
- Magic Move matches objects that appear on both slides
- Build Order: Animate › Build Order; set On Click, After Previous or With Previous
- Instant Alpha: select image › Format › Image › Instant Alpha, drag over the background
- Alignment guides appear while dragging; hold ⌘ to turn them off temporarily

AUTOMATE IT

- Magic Move between duplicated slides instead of hand-built motion paths
- Copy Animation / Paste Animation to repeat builds

BUILDING SHORTCUTS

```
⇧⌘N              # new slide
⌘D               # duplicate slide or object
⇧⌘H              # skip / unskip slide
⌥⌘G   ⇧⌥⌘G       # group / ungroup
⇧⌘F   ⇧⌘B        # bring to front / send to back
⌥⇧⌘F  ⌥⇧⌘B       # bring forward / send backward
⌘L    ⌥⌘L        # lock / unlock
⌥ + drag         # duplicate while dragging
⇧ + drag         # constrain movement / proportions
⌘ + drag         # turn off alignment guides
Format › Copy Animation / Paste Animation
```

CODE EXPLAINED

→ Duplicate the slide (⌘D) before making the Magic Move version

→ Skipped slides stay in the deck but don't play

→ Locked background objects can't be moved by accident

→ Builds can run by paragraph, so bullets appear one at a time

OTHER SUITES

⇄ **Microsoft 365** — PowerPoint Morph (same idea as Magic Move), Animation Pane

⇄ **Google Workspace** — Slides animations and transitions; no Magic Move

PLATFORM DIFFERENCES

- **Mac** — every build, Instant Alpha, Magic Move
- **iPad / iPhone** — builds and Magic Move; Apple Pencil drawing
- **iCloud (web)** — basic builds and transitions

BUILD TYPES

TYPE	EXAMPLE
Build In	Appear, Dissolve, Move In
Build Out	Disappear, Fade, Move Out
Action	Move, Rotate, Scale, Opacity
Magic Move	Transition matching objects
By paragraph / series	Bullets, chart series one at a time

BUILD ORDER OPTIONS

OPTION	PLAYS
On Click	Next click
After Previous	When the last ends
With Previous	Same time
Delay	Wait first
Duration	Speed

⚠ COMMON MISTAKES

- Too many builds per slide; they slow the talk
- Changing an object's text on the second slide, which breaks the Magic Move match
- Long durations; 0.5 s reads well

✓ TIPS & BEST PRACTICES

- Use Magic Move to "zoom" into part of a chart
- Lock background shapes before building animations
- Rehearse with the Build Order window open

QUICK SUMMARY

- Magic Move — duplicate, change, transition
- Build In / Out / Action — object motion
- Build Order — sequence and timing
- ⌘L — lock objects

Presenting & Exporting

RUN THE SHOW FROM THE KEYBOARD

Play (⌥⌘P) starts the show; the **presenter display** shows notes, the next slide and a timer on your screen. During a show, type a slide number and press Return to jump, press B or W to blank the screen, and X to swap displays if notes appear on the projector. **Rehearse Slideshow** practices timing; an **iPhone** with Keynote works as a remote.

NOTES

- Play › Customize Presenter Display to choose what you see
- Press `?` during a show to list every shortcut
- Play › Rehearse Slideshow for timed practice
- Keynote on iPhone / iPad: Keynote Remote controls the show over Wi-Fi
- Play in Window shows the deck in a window (useful for video calls)

DURING THE SLIDESHOW

```
⌥⌘P            # play from the current slide
→  Space  N    # next build / slide
←  P           # previous
12 ↩           # go to slide 12
Home / End     # first / last slide
B   W          # black / white screen
X              # swap presenter and audience displays
S              # show / hide slide number
?              # show all shortcuts
H              # hide the show (switch to another app)
Esc  /  Q      # end show
```

CODE EXPLAINED

- → Shortcuts are single keys while a show is running
- → Swap displays (`X`) fixes notes showing on the projector
- → Hidden show (`H`) lets you switch apps, then click Keynote to resume
- → Skipped slides can still be reached by number

OTHER SUITES

- ⇄ **Microsoft 365** — PowerPoint Presenter View, `F5`, number + Enter
- ⇄ **Google Workspace** — Slides Presenter view, `⌘↩` / `Ctrl+Enter`, Q&A

PLATFORM DIFFERENCES

- ▣ **Mac** — presenter display, rehearse, export movie / GIF
- ▣ **iPad / iPhone** — present, AirPlay, remote control
- ▣ **iCloud (web)** — play in browser

EXPORT OPTIONS

EXPORT	USE
PDF (with notes)	Handouts
PowerPoint	Office users
Movie	Recorded or timed video
Animated GIF	Short loops
Images	One per slide
HTML	Web viewing

BEFORE YOU PRESENT

CHECK	WHY
Presenter display set	Notes on your screen
Do Not Disturb / Focus	No pop-ups
Fonts embedded / local	Look stays the same
iPhone remote paired	Walk around
PDF backup	If the Mac fails

⚠ COMMON MISTAKES

- Mirroring displays so the audience sees your notes
- Clicking back through slides during Q&A; type the number
- Presenting from a file still downloading from iCloud

✓ TIPS & BEST PRACTICES

- Press `?` once before your talk to review shortcuts
- Use `B` to pause on a black screen for discussion
- Record a movie version for people who missed it

QUICK SUMMARY

- `⌥⌘P` — play
- `12 ↩` — jump to slide
- `B / W` — blank screen
- `X` — swap displays

Apple Mail Efficiency

A FASTER INBOX IN MAIL

Apple Mail handles iCloud, Gmail, Exchange and other accounts in one inbox. Keyboard shortcuts cover every action, **search tokens** narrow results, **Smart Mailboxes** save searches, and **VIPs** surface important people. **Send Later**, **Remind Me** and **Follow Up** keep things from slipping, and **categories** (macOS 15.4+) sort mail into Primary, Transactions, Updates and Promotions.

NOTES

- Smart Mailbox: Mailbox › New Smart Mailbox; rules like From is VIP and Unread
- VIP: hover a sender name › Add to VIPs
- Send Later: click and hold the Send button
- Remind Me: right-click a message › Remind Me
- Categories: View › Show Mail Categories (turn off for a single list)

AUTOMATE IT

- Smart Mailboxes instead of manual searches every morning
- VIPs + notifications only for VIPs

MAIL SHORTCUTS & SEARCH

```
⌘N              # new message
⌘R  ⇧⌘R         # reply / reply all
⇧⌘F             # forward
⇧⌘D             # send
⇧⌘N             # get new mail
⇧⌘U             # mark read / unread
⇧⌘L             # flag
^⌘A             # archive
⌥⌘F             # search
⌘1  ⌘2          # go to Inbox / VIPs (favorites order)

from:dana subject:invoice date:last week
attachment:pdf  is:flagged  is:unread
```

CODE EXPLAINED

- → Mail search turns typed words into tokens (From, Subject, Date) you can click to adjust
- → `⌘1` … jump to favorites in the Favorites bar order
- → Archive moves mail out of the inbox without deleting
- → Rules (sheet 21) can file mail automatically

OTHER SUITES

- ⇄ **Microsoft 365** — Outlook search operators, Focused Inbox, Quick Steps
- ⇄ **Google Workspace** — Gmail operators, labels, snooze, filters

PLATFORM DIFFERENCES

- ▣ **Mac** — Smart Mailboxes, rules, categories
- ▣ **iPad / iPhone** — categories, Remind Me, Send Later; no Smart Mailbox editing
- ▣ **iCloud (web)** — iCloud Mail only; rules via settings

SEARCH TOKENS

TYPE	FINDS
`from: / to:`	People
`subject:`	Subject words
`date:yesterday / last week`	Dates
`attachment:`	Attachment name / type
`is:flagged / is:unread`	State
`mailbox:`	Folder

INBOX TOOLS

TOOL	DOES
`Send Later`	Schedule a message
`Remind Me`	Bring a message back
`Follow Up`	Suggests replies that got none
`Hide My Email`	Unique iCloud address
`Mute thread`	Stop notifications
`Categories`	Primary / Transactions / Updates / Promotions

⚠ COMMON MISTAKES

- Leaving categories on and missing mail in Updates; check the setting
- Deep folder trees instead of Smart Mailboxes and search
- Using flags with no plan; pair flags with a Smart Mailbox

✓ TIPS & BEST PRACTICES

- Make a Smart Mailbox "Flagged and from VIPs"
- Undo Send: Settings › Composing › Undo send delay
- Use Hide My Email for sign-ups

QUICK SUMMARY

- `⇧⌘D` — send
- `from: subject: date:` — search tokens
- `Smart Mailboxes` — saved searches
- `Send Later / Remind Me` — timing tools

Rules, Signatures & Accounts

Mail rules & conditions · apply rules · signatures per account · Gmail / Exchange / iCloud · Mail Privacy Protection · Hide My Email · block & mute

LET MAIL DO THE FILING

Rules (Mail › Settings › Rules) act on incoming mail: move, flag, colour, forward, play a sound or run a Shortcut. **Signatures** are set per account. Mail handles **iCloud**, **Gmail**, **Exchange** and IMAP accounts side by side. **Mail Privacy Protection** hides your IP and blocks tracking pixels.

NOTES

- Rules run on the Mac when Mail is open; iCloud rules (iCloud.com) run on the server
- Apply rules to existing mail: select messages › Message › Apply Rules (⌥⌘L)
- Signatures: Settings › Signatures; drag to an account; choose a default
- Block a sender: hover the name › Block Contact
- Privacy: Settings › Privacy › Protect Mail Activity

AUTOMATE IT

- Rules that file newsletters and receipts as they arrive
- Rule action Run Shortcut for custom automation (sheet 24)

WHERE TO FIND IT

TASK	WHERE
Rules	Mail › Settings › Rules
Apply rules now	Message › Apply Rules (⌥⌘L)
Signatures	Settings › Signatures
Accounts	System Settings › Internet Accounts
Privacy	Settings › Privacy
Block / mute	Hover name / Message menu

CREATE A RULE

1. Mail › **Settings** › **Rules** › Add Rule
2. Name it (e.g. "Receipts")
3. Condition: **From contains** store@… or **Subject contains** receipt
4. Action: **Move Message** to mailbox Receipts; add Mark as Read
5. Click OK; choose **Apply** to run it on existing mail
6. For server-side rules on iCloud mail, use iCloud.com › Mail › Settings › Rules

OTHER SUITES

⇄ **Microsoft 365** — Outlook rules (server-side), Quick Steps, signatures

⇄ **Google Workspace** — Gmail filters from searches, signatures, vacation responder

PLATFORM DIFFERENCES

- **Mac** — every rule action (scripts, Shortcuts)
- **iPad / iPhone** — no rule editor; server rules from iCloud.com still apply
- **iCloud (web)** — server-side rules for iCloud addresses

USEFUL RULES

IF	THEN
From is in VIPs	Play sound, flag
Subject contains receipt	Move to Receipts
Any recipient contains list@	Move to Lists
Subject contains [Ticket]	Move to Tickets, mark read
Sender is not in Contacts	Colour message

ACCOUNT TYPES

ACCOUNT	NOTES
iCloud	Server rules, Hide My Email
Gmail	Labels appear as mailboxes
Exchange / Microsoft 365	Server rules set in Outlook
IMAP / POP	Rules run on the Mac

⚠ COMMON MISTAKES

- Relying on Mac rules while the Mac is asleep; use server rules for 24/7 filing
- Rules in the wrong order; Mail stops at the first rule that moves a message
- Using one signature for work and personal accounts

✓ TIPS & BEST PRACTICES

- Put specific rules first
- Use iCloud.com rules for iCloud addresses so iPhone stays tidy
- Add a plain-text signature for replies

QUICK SUMMARY

- `Settings › Rules` — automatic filing
- `⌥⌘L` — apply rules now
- `Signatures per account` — work vs personal
- `Protect Mail Activity` — privacy

Sheet 22

Calendar, Notes & Reminders

Calendar quick event · shared calendars · Notes tags & smart folders · Quick Note · Reminders lists, sections & smart lists · reminders from Mail

THE REST OF THE APPLE TOOLKIT

Calendar creates events from plain language ("Lunch with Sam Friday noon") and shares calendars through iCloud. **Notes** supports checklists, tables, **tags** and **smart folders**, and **Quick Note** captures from anywhere. **Reminders** organizes tasks into lists with sections, tags and **smart lists**, and can be created from Mail.

NOTES

- Calendar quick event: + button or `⌘N`, then type naturally
- Share an iCloud calendar: right-click it › Share Calendar
- Notes tags: type `#tag`; smart folders filter by tag, date, checklist
- Quick Note: hot corner or `fn Q` / Globe `Q`
- Reminders: drag a Mail message into Reminders to link it

SHORTCUTS

```
# Calendar
⌘N              # new event (natural language)
⌘T              # go to today
⌘1 ⌘2 ⌘3 ⌘4     # day / week / month / year
⌘→  ⌘←          # next / previous period

# Notes
⌘N              # new note
⇧⌘L             # checklist
⇧⌘U             # mark checklist item done
⌥⌘T             # insert table
fn Q            # Quick Note (Globe + Q)

# Reminders
⌘N              # new reminder
⇧⌘N             # new list
```

CODE EXPLAINED

→ Natural-language events understand dates, times and "every Monday"

→ Notes checklists can be sorted to move done items to the bottom

→ Quick Notes link to the app or web page you were in

→ Reminders smart lists filter by tag, date, location, priority

OTHER SUITES

⇄ **Microsoft 365** — Outlook Calendar, OneNote, To Do

⇄ **Google Workspace** — Google Calendar, Keep, Tasks

PLATFORM DIFFERENCES

- **Mac** — all three apps, Quick Note
- **iPad / iPhone** — same apps; Siri and widgets
- **iCloud (web)** — Calendar, Notes, Reminders in the browser

NOTES FEATURES

FEATURE	HOW
Tags	`#project` in the note
Smart folder	File › New Smart Folder
Checklist	⇧⌘L
Table	⌥⌘T
Share / collaborate	Share button
Lock note	Lock button (password / Touch ID)

REMINDERS FEATURES

FEATURE	USE
Sections	Group tasks in a list
Tags	Cross-list filtering
Smart lists	Saved filters
Assign (shared list)	Give tasks to people
Early reminder	Notify before due

⚠ COMMON MISTAKES

- Keeping to-dos in Notes, Reminders and Mail flags at once
- Sharing a calendar publicly when only family needed it
- Forgetting locked notes need their password on every device

✓ TIPS & BEST PRACTICES

- Use Notes tags + smart folders instead of nested folders
- Drag emails into Reminders for linked follow-ups
- Type events in plain language and let Calendar parse them

QUICK SUMMARY

- `⌘N` in Calendar — natural-language event
- `#tag` — Notes tags
- `fn Q` — Quick Note
- `Smart lists` — Reminders filters

AppleScript & JXA for iWork

SCRIPT THE APPS

Pages, Numbers and Keynote are scriptable with **AppleScript** or **JavaScript for Automation (JXA)** in **Script Editor**. Scripts can read and write table cells, loop through rows, change text, and batch-export files to PDF or Office formats. Save scripts to the **Script menu** for one-click use.

NOTES

- Script Editor: Applications › Utilities › Script Editor
- Dictionary: File › Open Dictionary › Numbers shows every command and object
- Pick JavaScript in the language pop-up for JXA
- Script menu: Script Editor › Settings › Show Script menu in menu bar
- First run asks for Automation permission (System Settings › Privacy & Security)

APPLESCRIPT: NUMBERS AND PAGES

```
-- Flag rows over 1000 in the front Numbers table
tell application "Numbers"
  tell table 1 of sheet 1 of front document
    repeat with r from 2 to row count
      set amt to value of cell 4 of row r
      if amt > 1000 then set value of cell 5 of row r to "Review"
    end repeat
  end tell
end tell

-- Export every open Pages document to PDF on the Desktop
tell application "Pages"
  repeat with d in documents
    set outPath to (path to desktop as text) & (name of d) & ".pdf"
    export d to file outPath as PDF
  end repeat
end tell
```

CODE EXPLAINED

- → Objects nest: document › sheet › table › row › cell
- → `row count` and `column count` give table size
- → `export … as PDF` also accepts `Microsoft Word`, `Microsoft Excel`, `Microsoft PowerPoint` formats
- → Run with `⌘R` in Script Editor; compile (check syntax) with `⌘K`

OTHER SUITES

- ⇄ **Microsoft 365** — VBA macros and Office Scripts
- ⇄ **Google Workspace** — Apps Script (JavaScript) in the browser

PLATFORM DIFFERENCES

- ▣ **Mac** — AppleScript and JXA only
- ▣ **iPad / iPhone** — no scripting; use Shortcuts (sheet 24)
- ▣ **iCloud (web)** — not scriptable

NUMBERS OBJECT MODEL

OBJECT	EXAMPLE
document	front document
sheet	sheet "Data"
table	table "Sales"
row / column	row 2, column "B"
cell	cell "D5", value of cell 4 of row r
range	range "A2:D20"

SAME IDEA IN JXA

APPLESCRIPT	JXA
tell application "Numbers"	const n = Application("Numbers")
front document	n.documents[0]
table 1 of sheet 1	.sheets[0].tables[0]
value of cell 4 of row r	.rows[r].cells[3].value()
set value of … to x	.value = x

⚠ COMMON MISTAKES

- Running scripts on the wrong front document; name documents explicitly
- Cell-by-cell loops over thousands of rows (slow); keep tables small or export CSV
- Denying the Automation prompt and wondering why nothing happens

✓ TIPS & BEST PRACTICES

- iWork has no macro recorder, so start from the Dictionary and examples
- Save scripts as apps or to the Script menu
- Use Shortcuts' Run AppleScript action to mix both

QUICK SUMMARY

- `Script Editor` — write and run
- `tell application "Numbers"` — target the app
- `export … as PDF` — batch export
- `JXA` — same in JavaScript

Shortcuts, Quick Actions & Text Replacements

Shortcuts app iWork actions · Quick Actions in Finder · Mail rule › run a Shortcut · Text Replacements · Automator (legacy) · Folder Actions

AUTOMATE WITHOUT CODE

The **Shortcuts** app strings actions together: convert files, make PDFs, add rows, send messages. Shortcuts can run from the menu bar, Finder **Quick Actions**, keyboard shortcuts, Siri, or a **Mail rule**. **Text Replacements** expand short codes everywhere. **Automator** still works but Shortcuts is the modern path.

NOTES

- Shortcut details (ⓘ) › Use as Quick Action / Pin in Menu Bar / Add Keyboard Shortcut
- Run AppleScript and Run Shell Script actions bridge to scripts
- Automations (iPhone / iPad, and Mac) run on time, arrival or other triggers
- Text Replacements: System Settings › Keyboard › Text Replacements
- Folder Actions (Automator) run when files land in a folder

AUTOMATE IT

- Quick Action "Make PDF" for any Pages / Keynote file in Finder
- Text Replacements for addresses, sign-offs and email templates

BUILD A FINDER QUICK ACTION

1. Open **Shortcuts** › New Shortcut
2. Add **Receive Files from Quick Actions** (input: Files)
3. Add **Make PDF** (or Convert Image, Rename Files…)
4. Add **Save File** to the same folder or a chosen one
5. Details (ⓘ) › tick **Use as Quick Action › Finder**
6. Right-click files in Finder › Quick Actions › your shortcut

WHERE TO FIND IT

TASK	WHERE
`Shortcuts app`	Applications › Shortcuts
`Quick Actions`	Finder › right-click › Quick Actions
`Keyboard shortcut`	Shortcut ⓘ › Add Keyboard Shortcut
`Mail rule action`	Mail rules › Run Shortcut
`Text Replacements`	System Settings › Keyboard
`Folder Actions`	Right-click folder › Folder Actions Setup

OTHER SUITES

- ⇄ **Microsoft 365** — Power Automate (cloud), Office Scripts, VBA
- ⇄ **Google Workspace** — Apps Script triggers, Workspace Flows

PLATFORM DIFFERENCES

- ▣ **Mac** — Shortcuts, Quick Actions, Automator, Folder Actions
- ▣ **iPad / iPhone** — Shortcuts and automations; no Automator
- ▣ **iCloud (web)** — no automation

HANDY ACTIONS

ACTION	DOES
`Make PDF`	Any document to PDF
`Convert Image`	HEIC → JPEG etc.
`Rename Files`	Batch rename
`Get Contents of URL`	Call a web API
`Run AppleScript / Shell Script`	Anything scripts can do
`Send Email / Message`	Notify people

WAYS TO RUN

TRIGGER	EXAMPLE
`Menu bar`	Pinned shortcut
`Keyboard`	^⌥⌘P
`Finder Quick Action`	Right-click files
`Siri`	"Make PDF"
`Automation`	Every weekday 8am
`Mail rule`	Run on matching mail

⚠ COMMON MISTAKES

- Text Replacement codes that are real words
- Building in Automator when Shortcuts covers it (Automator gets few updates)
- Shortcuts that overwrite originals; save output to a new file

✓ TIPS & BEST PRACTICES

- Start from the Gallery tab for ready-made shortcuts
- Pin daily shortcuts to the menu bar
- Combine Shortcuts with AppleScript for iWork-specific steps

QUICK SUMMARY

- `Shortcuts` — no-code automation
- `Quick Actions` — right-click in Finder
- `Text Replacements` — expand codes
- `Run AppleScript` — bridge to scripts

Sheet 25

Apple Intelligence & Creator Studio Features

Writing Tools · Mail summaries & smart replies · Image Playground / Image Wand · ChatGPT extension · Creator Studio features · verify AI edits

AI IN APPLE'S APPS

On supported Macs, iPads and iPhones, **Apple Intelligence** adds **Writing Tools** (proofread, rewrite, summarise) in Pages, Keynote, Mail and any text field, **summaries** and **smart replies** in Mail, and **Image Playground** for generated images. An optional **ChatGPT extension** can handle requests Apple's models can't. **Apple Creator Studio** adds more intelligent features to Pages, Numbers and Keynote by subscription.

NOTES

- Turn on: System Settings › Apple Intelligence & Siri
- Writing Tools: select text › right-click › Writing Tools (or the ✦ icon)
- Mail: summary at the top of long threads; Priority messages; Smart Reply
- Image Playground and Image Wand create images from prompts or sketches
- Feature availability varies by device, language, region and subscription

PROMPTS & WRITING TOOLS

```
# Writing Tools (select text first)
Proofread                  # fixes grammar, shows changes
Rewrite  /  Friendly  /  Professional  /  Concise
Summary  /  Key Points  /  Table  /  List

# Describe your change (free-form)
"Make this sound warmer and cut it to 80 words"
"Turn these notes into a bulleted agenda"

# Keynote / Pages with Creator Studio (check current features)
"Draft speaker notes for this slide, 45 seconds"
"Suggest a layout for this page"
```

CODE EXPLAINED

- → Proofread marks each change so you can review them one by one
- → Writing Tools work in any app with text fields, not just iWork
- → ChatGPT requests ask before sending anything off-device
- → Creator Studio features change over time; check Apple's current list

OTHER SUITES

- ⇄ **Microsoft 365** — Copilot in Word, Excel, PowerPoint, Outlook
- ⇄ **Google Workspace** — Gemini side panel, Help me write, `AI()` in Sheets

PLATFORM DIFFERENCES

- ▣ **Mac** — Apple silicon Macs with Apple Intelligence
- ▣ **iPad / iPhone** — supported models only
- ▣ **iCloud (web)** — no Apple Intelligence features

WHERE IT SHOWS UP

APP	FEATURE
Pages / Keynote	Writing Tools, Image Playground
Mail	Summaries, Priority, Smart Reply
Notes	Writing Tools, summaries of recordings
Any text field	Writing Tools
Photos / Notes	Image Wand (sketch → image)

VERIFY BEFORE USING

OUTPUT	CHECK
Rewrites	Meaning unchanged, names right
Summaries	Nothing important dropped
Smart replies	Tone and commitments
Generated images	Accuracy, appropriateness
Numbers analysis	Hand-check calculations

⚠ COMMON MISTAKES

- Accepting a rewrite without reading it
- Trusting a Mail summary instead of reading important threads
- Assuming every device has the same features

✓ TIPS & BEST PRACTICES

- Use Proofread as a last pass on every document
- Use "Describe your change" for specific edits
- See the Agentic Coding set for prompting techniques

QUICK SUMMARY

- `Writing Tools` — proofread, rewrite, summarise
- `Mail summaries` — long threads
- `Image Playground` — generated images
- `Verify` — every AI result

Quick Ref — Pages & Keynote

shared editing · text & styles · objects · Pages · Keynote · presenting

SHARED EDITING (02)

ACTION	MAC	WHERE / NOTE
Copy / paste style	⌥⌘C / ⌥⌘V	Copy formatting
Paste and match style	⌥⇧⌘V	Paste plain
Bold / italic / underline	⌘B / ⌘I / ⌘U	
Bigger / smaller font	⌘+ / ⌘−	
Align left / center / right	⌘{ / ⌘| / ⌘}	
Super / subscript	^⌘+ / ^⌘−	
Fonts / Colors panel	⌘T / ⇧⌘C	System panels
Find / next / use selection	⌘F / ⌘G / ⌘E	
Add link	⌘K	URL, email, bookmark, slide
Add comment	⇧⌘K	
Rulers	⌘R	
Help menu search	⇧⌘/	Find any command

OBJECTS & ARRANGE (05, 18)

ACTION	MAC	WHERE / NOTE
Group / ungroup	⌥⌘G / ⇧⌥⌘G	
Front / back	⇧⌘F / ⇧⌘B	
Forward / backward	⌥⇧⌘F / ⌥⇧⌘B	
Lock / unlock	⌘L / ⌥⌘L	
Duplicate	⌘D	or ⌥ + drag
Constrain	⇧ + drag	Straight lines, keep ratio
Guides off while dragging	⌘ + drag	
Instant Alpha	Format › Image	Remove background

PAGES (05–08)

ACTION	MAC	WHERE / NOTE
Show invisibles	⇧⌘I	¶ and breaks
Word count	⇧⌘W	
Track Changes	⇧⌘E	
Highlight	⇧⌘H	
Style shortcuts	F1–F8	Assign in style menu
Next misspelling	⌘;	
Spelling panel	⇧⌘;	
Page break	Insert › Page Break	
Table of contents	Insert › Table of Contents	
Mail merge	Document › Mail Merge	

KEYNOTE (17–19)

ACTION	MAC	WHERE / NOTE
New slide	⇧⌘N	
Skip slide	⇧⌘H	
Edit masters	View › Edit Master Slides	
Play	⌥⌘P	From current slide
Go to slide n	n ↵	During show
Black / white	B / W	During show
Swap displays	X	During show
Slide number	S	During show
Shortcut list	?	During show
End show	Esc / Q	

TEXT MOVEMENT (02)

ACTION	MAC	WHERE / NOTE
Word left / right	⌥← / ⌥→	
Line start / end	⌘← / ⌘→	
Document start / end	⌘↑ / ⌘↓	
Select by word	⌥⇧← / ⌥⇧→	
Select to line end	⌘⇧→	
Delete word	⌥⌫	
Undo / redo	⌘Z / ⇧⌘Z	

FILES & SHARING (01, 03, 04)

ACTION	MAC	WHERE / NOTE
New / open	⌘N / ⌘O	Template chooser on ⌘N
Duplicate (Save As)	⇧⌘S	Hold ⌥ for Save As
Export	File › Export To	Word, PowerPoint, PDF, EPUB
Collaborate	Share button	Invite or link
Browse versions	File › Revert To	Browse All Versions
Save as template	File › Save as Template	
Set password	File › Set Password	

Quick Ref — Numbers & Mail

NUMBERS NAVIGATION (09)

ACTION	MAC	WHERE / NOTE
Next / previous cell	Tab / ⇧Tab	Tab at end adds column
Next / previous row	↩ / ⇧↩	↩ at end adds row
Jump to table edge	⌘ + arrow	
Extend to edge	⇧⌘ + arrow	
Stop editing	Esc	Keeps cell selected
Line break in cell	⌥↩	
Copy / paste style	⌥⌘C / ⌥⌘V	Cell formatting
Paste and match style	⌥⇧⌘V	

ROWS, COLUMNS & ENTRY (09–10)

ACTION	MAC	WHERE / NOTE
Add row above / below	⌥↑ / ⌥↓	
Add column before / after	⌥← / ⌥→	
Delete rows / columns	⌥⌘⌫	
Paste formula results	⇧⌘V	Values only
Fill down / right	Table › Autofill Cells	or drag yellow handle
Start a formula	=	
Quick calculations	Select cells	Sum / average at bottom
Functions Browser	Type =	Right side

TABLES & ORGANIZE (11–15)

ACTION	MAC	WHERE / NOTE
Sort / filter / categories	Organize button	
Pivot table	Organize › Create Pivot Table	
Refresh pivot	Organize › Refresh	
Cell format	Format › Cell › Data Format	
Highlight rules	Format › Cell › Conditional Highlighting	
Chart	Chart button	
Preserve row / column	Reference token › Preserve	Like $

MAIL (20–21)

ACTION	MAC	WHERE / NOTE
New / reply / reply all	⌘N / ⌘R / ⇧⌘R	
Forward / send	⇧⌘F / ⇧⌘D	
Get mail	⇧⌘N	
Read / unread	⇧⌘U	
Flag / archive	⇧⌘L / ^⌘A	
Search	⌥⌘F	Tokens: from: subject:
Apply rules	⌥⌘L	
Send later	Hold Send	

CALENDAR, NOTES & REMINDERS (22)

ACTION	MAC	WHERE / NOTE
New event	⌘N	Natural language
Today	⌘T	
Day / week / month / year	⌘1 / ⌘2 / ⌘3 / ⌘4	
Notes checklist	⇧⌘L	
Notes table	⌥⌘T	
Quick Note	fn Q	Globe + Q
Mark checklist done	⇧⌘U	
New reminder / list	⌘N / ⇧⌘N	Reminders

AUTOMATION (23–24)

ACTION	MAC	WHERE / NOTE
Run / compile script	⌘R / ⌘K	Script Editor
Open dictionary	File › Open Dictionary	
Quick Action	Finder › right-click	
Shortcut keyboard key	Shortcut ⓘ	
Text Replacements	System Settings › Keyboard	
Writing Tools	Right-click selection	Apple Intelligence
Script menu	Script Editor › Settings	Show in menu bar
Automation permission	Privacy & Security	Automation

Sheet 28

Quick Ref — Numbers Formulas

references · logic · math · lookup · arrays · text & regex · date & duration

REFERENCES (11)

FORMULA	DOES
`B2 / B2:B20`	Address
`Amount`	Header-name column
`Price Widget`	Column + row headers
`Sales::Amount`	Other table
`Q1::Sales::B2`	Other sheet
`$B$2`	Preserved
`× ÷ ≥ ≤ ≠`	Display for `* / >= <= <>`

LOGIC & MATH (13)

FORMULA	DOES
`IF(t, yes, no)`	Choose
`IFS(t1,v1, …, TRUE,else)`	First true
`SWITCH(x, a,"A", other)`	Match
`IFERROR(x, alt)`	Catch errors
`SUMIFS(sum, rng,crit, …)`	Conditional sum
`COUNTIFS(rng,crit, …)`	Conditional count
`AVERAGEIFS / MAXIFS / MINIFS`	Conditional stats
`SUMIFS(Amt, Name, REGEX("^A"))`	Pattern criteria

LOOKUP & ARRAYS (13)

FORMULA	DOES
`XLOOKUP(v, look, ret, "none")`	Exact, with miss value
`XLOOKUP(v, look, ret, , −1)`	Next smaller
`XMATCH(v, look)`	Position
`INDEX(ret, MATCH(v, look, 0))`	Classic
`VLOOKUP(v, rng, col, FALSE)`	Legacy
`FILTER(rng, cond)`	Matching rows
`SORT(rng) / UNIQUE(rng)`	Sort / distinct
`TAKE / VSTACK / TOCOL`	Reshape
`LET(n, v, calc) / LAMBDA`	Named steps / functions
`GETPIVOTDATA(…)`	Value from a pivot

TEXT & REGEX (14)

FORMULA	DOES
`TEXTBEFORE(t, "@") / TEXTAFTER`	Part of text
`TEXTBETWEEN(t, "(", ")")`	Between markers
`TEXTSPLIT(t, ",")`	Split
`TEXTJOIN(", ", TRUE, rng)`	Join, skip blanks
`TRIM / PROPER / PLAINTEXT`	Tidy
`REGEX.EXTRACT(t, "\d+")`	Pull a match
`COUNTMATCHES(t, REGEX("x"))`	Count matches
`SUBSTITUTE(t, REGEX("\s+"), " ")`	Regex replace

DATE & DURATION (14)

FORMULA	DOES
`TODAY() / NOW()`	Date / date + time
`EOMONTH(d, 0)`	Month end
`EDATE(d, 3)`	+3 months
`NETWORKDAYS(a, b) / WORKDAY`	Business days
`DATEDIF(a, b, "Y")`	Whole years
`DAYNAME(d) / MONTHNAME(m)`	Names
`DUR2HOURS(dur)`	Duration → hours
`DURATION(0, 0, 2, 30)`	2h 30m

ERRORS (11)

MESSAGE	FIX
`Invalid reference`	Check renamed / deleted target
`Divide by zero`	Guard with IFERROR
`Type mismatch`	Text where number expected
`Circular reference`	Formula uses itself
`Syntax error`	Parentheses / arguments
`Function not found`	Not in this Numbers version